



Part One: Choose the correct answer

(10 marks)

1	2	3	4	5

1- One of the following matrices is in R.R.E form

(A) $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

B) $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

C) $\begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

D) $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

2- The corresponding linear system for the augmented matrix $\begin{bmatrix} 2 & 1 & 3 & -5 \\ 1 & 2 & 4 & 0 \\ -1 & 6 & 2 & 1 \end{bmatrix}$ is:

(A) $2X_1 + X_2 + 3X_3 + 5 = 0$

B) $2X_1 + X_2 + 3X_3 = 5$

$X_1 + 2X_2 + 4X_3 = 0$

$X_1 + 2X_2 + 4X_3 = 0$

$-X_1 + 6X_2 + 2X_3 - 1 = 0$

$-X_1 + 6X_2 + 2X_3 = 1$

C) $2X_1 + X_2 + 3X_3 + 5 = 0$

D) $2X_1 + X_2 + 3X_3 + 5 = 0$

$X_1 + 2X_2 + 4X_3 = 0$

$X_1 + 2X_2 + 4X_3 = 0$

$-X_1 + 6X_2 + 2X_3 - 1 = 0$

$X_1 - 6X_2 - 2X_3 - 1 = 0$

3- The solution of the linear system with the corresponding augmented matrix

$\begin{bmatrix} 2 & 5 & 8 \\ 4 & 10 & 16 \end{bmatrix}$ is:

A) The system has no solution

B) $X_1 = 4$ and $X_2 = 0$

(C) $X_1 = 4 - \frac{5t}{2}$ and $X_2 = t; t \in \mathbb{R}$

D) $X_1 = 8 - 5t$ and $X_2 = t; t \in \mathbb{R}$

4- One of the following statements is false for any symmetric matrix A

A) $tr(2A) = tr(A) + tr(A^T)$.

B) AA^T is symmetric.

C) A is a square matrix.

(D) The equation $AX = B$ has only one solution.

5- Let A be a matrix with $\det(A) = 7$, and B is the matrix result by multiplying the third row in A by the scalar $\alpha = 3$ and the second column in A by the scalar $\beta = -2$ then $\det(B) =$

A) $\frac{-7}{6}$

(B) -42

C) 21

D) -14

Part two: Fill the blank. (10 marks)

1- Let $A = \begin{bmatrix} 2 & -7 \\ -3 & 0 \end{bmatrix}$ then $A^{-1} = \frac{1}{2 \cdot 1} \begin{bmatrix} 0 & 7 \\ 3 & 2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & 7 \\ 3 & 2 \end{bmatrix} = A^{-1}$

2- Let $B = \begin{bmatrix} 5 & 0 & 2 \\ 0 & 3 & 7 \\ 0 & 0 & b \end{bmatrix}$ with $\det(B) = 10$ then $b = \frac{10 - \frac{2}{5}}{5} = \frac{10 - \frac{2}{5}}{5}$

3- If $A^{-1} = \begin{bmatrix} 1 & 5 & 3 \\ 0 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix}$ then the solution of the equation $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ is $\begin{bmatrix} 20 \\ 7 \\ 1 \end{bmatrix}$

4- If A is a square matrix satisfies $A^2 + 5A + 6I = I$, then $(A + 3I)^{-1} = \dots\dots\dots 5$

5- Let A and B be two symmetric matrix with the same size, then AB is symmetric matrix iff $AB = \dots\dots B.A.$

Part Three: Essay Questions

1- Use Cramer's Rule to solve the linear system

$$\begin{aligned} x + 2y - z &= 5 \\ 2x + y + z &= 0 \\ -x + y + 2z &= 1 \end{aligned}$$

(4 marks)

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow \Delta_1$$

$$\Delta = 1(2 - (1 \times 1)) - 2(4 + 1) - 1(2 + 1)$$

$$= 1 - 10 - 3 = -12$$

$$\Delta_1 = 5(2 - 1) - 2(0 - 1) - 1(0 - 1) = 5 + 2 + 1 = 8$$

$$\Delta_2 = 1(0 - 2) - 5(4 + 1) - 1(4 - 0)$$

$$= -2 - 25 - 4 = -31$$

$$x = -\frac{8}{12}, y = \frac{31}{12}, z = -1$$

$$\Delta_3 = 1 + 10 + 0 - (-5 + 4 + 0) = 11 + 1 = 12$$

2- Use Gauss-Jordan Method to solve the system

(2 marks)

$$\begin{aligned} x + 3y &= 5 \\ 2x + 4y &= 8 \end{aligned}$$

$$\begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 8 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & 5 \\ 0 & -2 & -2 \end{bmatrix} \xrightarrow{-2R_1 + R_2}$$

$$\Rightarrow \begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \times -2}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{-3R_2 + R_1}$$

$$(x, y) = (2, 1)$$

3- Find the inverse matrix for $A = \begin{bmatrix} 1 & 3 & 2 \\ -2 & 1 & 4 \\ 0 & 5 & 0 \end{bmatrix}$

(4 marks)

$$A^{-1} = \frac{1}{\det} \text{adj}(A)$$

$$= \frac{-1}{40} \begin{bmatrix} -20 & 20 & 0 \\ 0 & 0 & 40 \\ -20 & 20 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & -1 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} = A^{-1}$$

$$MA = \begin{bmatrix} -20 & 0 & -20 \\ -20 & 0 & -20 \\ 0 & -40 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} -20 & 0 & -20 \\ 20 & 0 & 20 \\ 0 & 40 & 0 \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} -20 & 20 & 0 \\ 0 & 0 & 40 \\ -20 & 20 & 0 \end{bmatrix}$$

$$\det = 1(-20) + 3(0) + 2(-10) = -20 - 20 = -40$$